Economics 5040: Dynamic Stochastic Modeling
Lecture 2: Intertemporal Production and General Equilibrium

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1 Intertemporal Production

In the previous lecture, we treated consumption today and consumption next year as two different commodities. The gross real interest rate R was the relative price of these two commodities. We now introduce production and ask a natural question: How can an economy transform goods available today into goods available tomorrow?

Consider a two-period economy. There is one physical good. At date $t = 0$, the economy is endowed with y_0 units of this good. The good can either be consumed immediately or invested as capital.

Let c_0 denote current consumption and k_0 denote investment. Feasibility at date 0 requires

$$c_0 + k_0 = y_0.$$

Investment produces output next year according to the production function

$$y_1 = f(k_0),$$

where we assume

$$f'(k_0) > 0 \qquad \text{and} \qquad f''(k_0) < 0.$$

Thus additional investment produces additional future output, but there are diminishing marginal returns to investment.

Since the economy ends at date 1, there is no reason to save any of date-1 output. Hence

$$c_1 = y_1 = f(k_0).$$

The economy faces a simple choice: consuming more today requires investing less today and therefore consuming less tomorrow.

2 The Intertemporal Production Possibility Frontier

Since

$$k_0 = y_0 - c_0,$$

we can substitute into the production function to obtain

$$c_1 = f(y_0 - c_0).$$

This equation describes the economy's *intertemporal production possibility frontier*. It shows all technologically feasible combinations of current and future consumption.

Differentiating gives

$$\frac{dc_1}{dc_0} = -f'(k_0).$$

Thus the absolute value of the slope of the production possibility frontier is

$$f'(k_0).$$

This is the *marginal rate of transformation* between current and future consumption.

The interpretation is straightforward. If we give up one unit of consumption today, we can invest approximately one additional unit. That additional investment produces approximately $f'(k_0)$ additional units of consumption next year.

Thus technology itself determines a rate at which current consumption can be transformed into future consumption.

3 The Planner's Problem

We now ask how much the economy *should* consume today and how much it should invest for tomorrow.

To answer this question, imagine a benevolent social planner who controls the economy's resources. In this simple economy there is only one representative consumer, so the planner chooses the allocation that maximizes this consumer's utility.

Suppose preferences are time separable:

$$U(c_0, c_1) = u(c_0) + \beta u(c_1),$$

where

$$u'(c) > 0, \qquad u''(c) < 0,$$

and $0 < \beta \leq 1$.

It is useful to state the planner's problem in its most explicit form. The planner chooses current consumption, future consumption, and investment:

$$\max_{c_0, c_1, k_0} \{u(c_0) + \beta u(c_1)\}$$

subject to

$$c_0 + k_0 \leq y_0,$$

$$c_1 \leq f(k_0),$$

and

$$c_0, c_1, k_0 \geq 0.$$

The first constraint says that current output can be divided between current consumption and investment. The second says that future consumption cannot exceed the output produced by today's investment.

Because utility is increasing in consumption, neither resource constraint will be slack at an optimum. Therefore,

$$c_0 + k_0 = y_0$$

and

$$c_1 = f(k_0).$$

We can consequently substitute the constraints into the objective function and write the planner's problem using only one choice variable:

$$\max_{0 \leq k_0 \leq y_0} \{u(y_0 - k_0) + \beta u(f(k_0))\}.$$

This formulation makes the economic tradeoff especially clear. Increasing k_0 reduces consumption today but increases consumption tomorrow.

4 The Optimal Intertemporal Allocation

For an interior solution, the first-order condition for the planner's problem is

$$-u'(y_0 - k_0) + \beta u'(f(k_0))f'(k_0) = 0.$$

Since

$$c_0 = y_0 - k_0 \qquad \text{and} \qquad c_1 = f(k_0),$$

we can write this condition as

$$u'(c_0) = \beta u'(c_1)f'(k_0).$$

Rearranging,

$$\boxed{\frac{u'(c_0)}{\beta u'(c_1)} = f'(k_0).}$$

The left-hand side is the consumer's marginal rate of substitution between current and future consumption. The right-hand side is the economy's marginal rate of transformation between current and future consumption.

Thus an efficient allocation satisfies

$$\boxed{MRS = MRT.}$$

The intuition is important. Suppose instead that

$$MRS > MRT.$$

The consumer values an additional unit of current consumption, relative to future consumption, more highly than the technology requires in order to produce that future consumption. The economy should invest less and consume more today.

If

$$MRS < MRT,$$

the technological return to sacrificing current consumption is greater than the consumer's subjective tradeoff. The economy should invest more.

Only when $MRS = MRT$ is there no further reallocation between current and future consumption that can make the consumer better off.

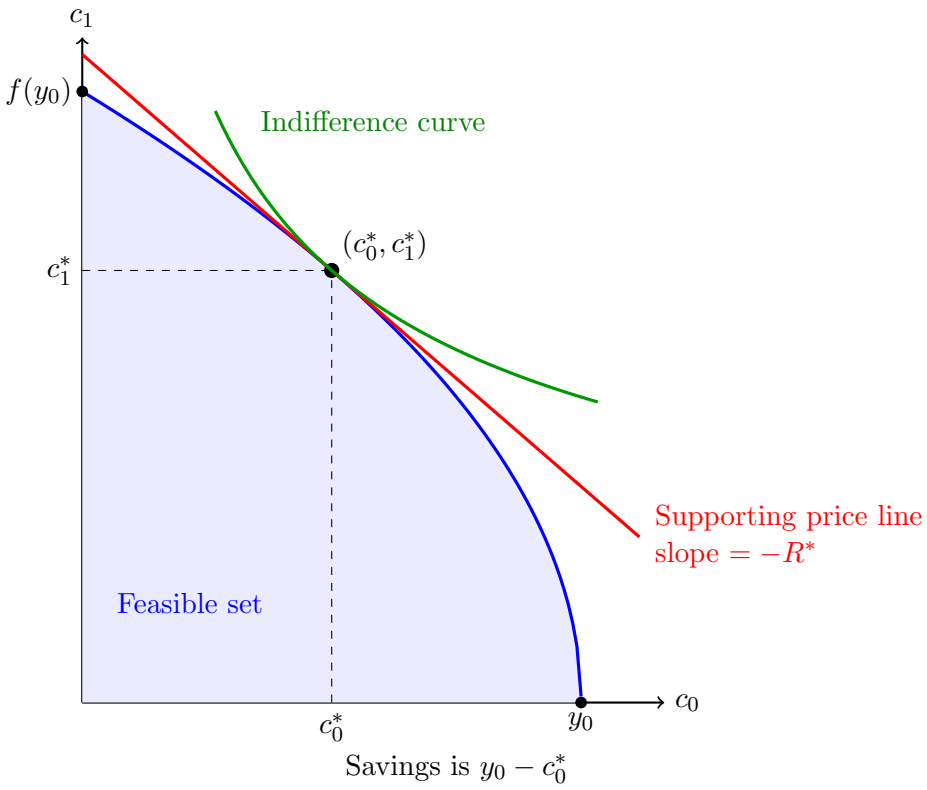


Figure 1: The planner chooses the point (c_0^*, c_1^*) at which an indifference curve is tangent to the intertemporal production possibility frontier. The supporting price line has slope $-R^*$. At the optimum, $MRS = R^* = MRT$.

5 Markets and the Gross Real Interest Rate

The social planner is a useful analytical device, but actual market economies are decentralized. Consumers make consumption and saving decisions, while firms make production and investment decisions.

Suppose that the gross real interest rate is R.

From the previous lecture, a utility-maximizing consumer who chooses strictly positive consumption in both periods satisfies

$$\frac{u'(c_0)}{\beta u'(c_1)} = R.$$

Thus

$$MRS = R.$$

Now consider a competitive firm. The firm can acquire one unit of the good today, invest it, and produce goods tomorrow. The present value of the firm's output is

$$\frac{f(k_0)}{R}.$$

If investment costs one unit of current consumption per unit of capital, the firm's present-value profit is

$$\Pi(k_0) = \frac{f(k_0)}{R} - k_0.$$

The firm chooses investment to solve

$$\max_{k_0 \geq 0} \left\{ \frac{f(k_0)}{R} - k_0 \right\}.$$

For an interior solution, the first-order condition is

$$\frac{f'(k_0)}{R} - 1 = 0,$$

or

$$\boxed{f'(k_0) = R.}$$

Thus

$$MRT = R.$$

Combining the consumer's optimality condition with the firm's optimality condition gives

$$\boxed{MRS = R = MRT.}$$

The gross real interest rate coordinates the decisions of consumers and firms. Consumers compare the market return on saving with their subjective willingness to substitute consumption across time. Firms compare the same market return with the productivity of investment.

In equilibrium, one price coordinates preferences and technology.

6 A First Look at the First Welfare Theorem

We have now solved the economy in two apparently different ways.

First, we imagined a social planner who directly chose consumption and investment. The planner's allocation satisfied

$$MRS = MRT.$$

Second, we allowed consumers and firms to make their own decisions, taking the gross real interest rate R as given. Their separate optimization problems gave

$$MRS = R$$

and

$$R = MRT.$$

Hence the competitive allocation also satisfies

$$MRS = MRT.$$

This is our first glimpse of one of the central results of general equilibrium theory: the *First Welfare Theorem*.

Loosely stated, the First Welfare Theorem says:

> Under appropriate conditions, a competitive equilibrium allocation is Pareto efficient.

An allocation is *Pareto efficient* if there is no other feasible allocation that makes at least one person better off without making someone else worse off.

In our economy there is only one consumer, so Pareto efficiency has a particularly simple meaning: there is no other technologically feasible consumption plan (c_0, c_1) that the consumer prefers.

The important idea is not that a planner actually directs the competitive economy. No planner tells the consumer how much to save, and no planner tells the firm how much to invest. Each takes prices as given and pursues its own objective.

Nevertheless, the market price R coordinates these independent decisions:

$$\underbrace{\frac{u'(c_0)}{\beta u'(c_1)}}_{\text{preferences}} = \underbrace{R}_{\text{market price}} = \underbrace{f'(k_0)}_{\text{technology}}.$$

This is the fundamental insight behind the First Welfare Theorem. Under the appropriate assumptions, competitive prices can decentralize an efficient allocation.

Later we will study economies with many consumers, many goods, uncertainty, and incomplete markets. We will then have to be much more careful about the conditions under which welfare theorems apply. But the basic economic logic is already visible in this simple two-period economy.